

Module Summary Report for Wine Quality Analysis

Overview

This project developed a reproducible data analysis workflow using the Wine Quality datasets from the UCI Machine Learning Repository. The workflow included data ingestion, cleaning, exploratory data analysis (EDA), visualization, and interpretation using Python and Jupyter Notebook. The analysis focused on identifying relationships between physicochemical wine properties and expert-assigned quality ratings while following reproducible data science practices.

Dataset Source:

Cortez, P., Cerdeira, A., Almeida, F., Matos, T., & Reis, J. (2009). Wine Quality [Dataset]. UCI Machine Learning Repository. <https://archive.ics.uci.edu/dataset/186/wine+quality>.

Dataset Description

The Wine Quality dataset contains physicochemical measurements(continuous) and expert quality ratings(integer) for Portuguese “Vinho Verde wines” (Cortez et al., 2009). Two datasets were used in this analysis: red wine samples and white wine samples. After removing duplicate observations and merging the datasets, the final dataset contained 5,320 observations and 13 variables, including the newly created wine_type identifier used to distinguish red and white wines.

Key variables analyzed included density, alcohol content, sulfur dioxide measurements, and the target variable, quality. Wine quality scores ranged from 0 to 10 and were assigned by wine experts through sensory evaluation.

Workflow Description

Data Ingestion

The red and white wine datasets were loaded using Pandas and inspected to verify successful ingestion, data types, and dataset structure. Initial inspection confirmed that all variables contained numeric values and that no missing data were present.

Data Cleaning

The data cleaning phase focused on removing duplicate records and preparing both datasets for merging. Duplicate observations were removed to prevent repeated records from influencing summary statistics and visualizations. A new `wine_type` column was added before merging to be able to distinguish between wine colour for analysis, and to preserve the original data structure.

Data preprocessing is an important stage of a reproducible data science workflow because it helps prepare and organize data prior to analysis (Danchev, 2022). Proper data cleaning also improves analytical reliability by reducing the influence of data anomalies on subsequent results.

Exploratory Data Analysis

Exploratory Data Analysis began with summary statistics, grouped analysis, and correlation analysis. Summary statistics provided an initial understanding of central tendency, dispersion, and the overall characteristics of the dataset before more detailed analysis was performed. Summary statistics are widely recognized as a useful first step in exploratory analysis because they provide a concise overview of the data and can help identify unusual observations and potential patterns (Agarwal et al., 2024).

The exploratory analysis focused on:

- Understanding the overall distribution of wine quality scores.
- Comparing physicochemical properties between red and white wines.
- Investigating relationships between physicochemical measurements and quality.

Visualizations

Four visualizations were developed to support the exploratory analysis:

Figure 1: Distribution of Wine Quality by Wine Type

Figure 2: Alcohol Content by Quality Score and Wine Type

Figure 3: Density by Quality Score and Wine Type

Figure 4: Correlation Heatmap of Wine Features

Boxplots were selected for Figures 2 and 3 because they provide an effective visual summary of distributions, medians, interquartile ranges, and outliers while supporting comparisons across multiple groups (Shreffler & Huecker, 2020).

Summary

The notebook concludes with a summary of findings, limitations, assumptions, and recommendations for future work.

Key Decisions and Assumptions

Cleaning Decisions

Duplicate records were removed from both datasets prior to analysis. Duplicate observations can influence descriptive statistics and potentially distort visualizations if retained. Because the dataset contained no missing values and all variables contained appropriate data types, no additional cleaning steps were taken. New feature 'wine_type' was added for dataset preservation and grouped analysis purposes.

Exploratory Analysis Focus

The analysis focused on summary statistics, grouped comparisons, and correlation analysis. This approach was chosen because it directly supported the project's research question:

How do physicochemical properties differ between red and white Vinho Verde wines, and which properties appear most associated with expert quality ratings?

The summary statistics provided a high-level statistical overview of the data, while grouped analysis and correlation analysis helped investigate relationships between wine characteristics and quality ratings as well as other features.

Visualization Design Decisions

Each visualization was designed to answer a specific analytical question and color palettes was chosen taking color blindness into consideration (Okabe-Ito Palette):

- **Figure 1** was created to examine the distribution of quality ratings between wine types.
- **Figure 2** was created to investigate the relationship between alcohol content and wine quality.

- **Figure 3** was created to investigate the relationship between density and wine quality.
 - An outlier was retained in the dataset but omitted from the boxplot visualization using `showfliers=False` to improve readability while preserving the underlying data for analysis
- **Figure 4** provided an overall view of interactions and relationships among all measured variables include the target.

Results and Interpretation

The analysis revealed that red and white wines possess noticeably different physicochemical compositions despite receiving similar average quality ratings. White wines exhibited higher average levels of residual sugar and sulfur dioxide, while red wines exhibited higher average levels of acidity and sulphates.

Summary statistics indicated that most wines received quality ratings between 5 and 6, with relatively few observations occurring at the highest and lowest quality levels. This suggests that average-quality wines dominate the dataset.

Figure 1 demonstrated that both wine types share similar quality distributions, although white wines exhibited slightly more observations with quality ratings above 6. A small number of white wines achieved the highest quality score of 9, while no red wines received this rating.

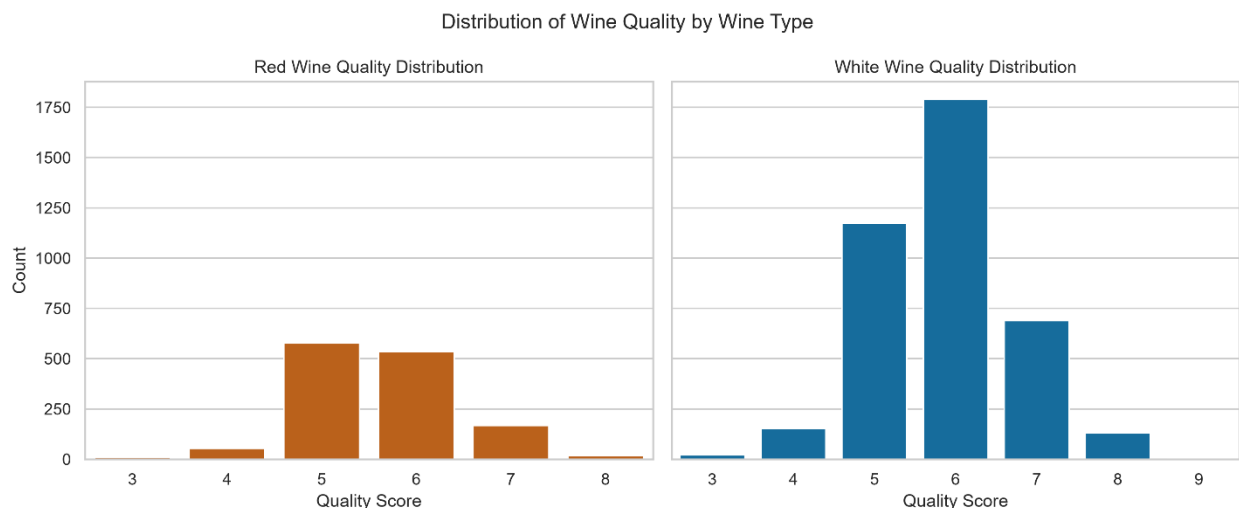


Figure 2 showed a clear positive relationship between alcohol content and wine quality. Higher-quality wines generally exhibited greater median alcohol concentrations, particularly among wines with quality scores above 6.

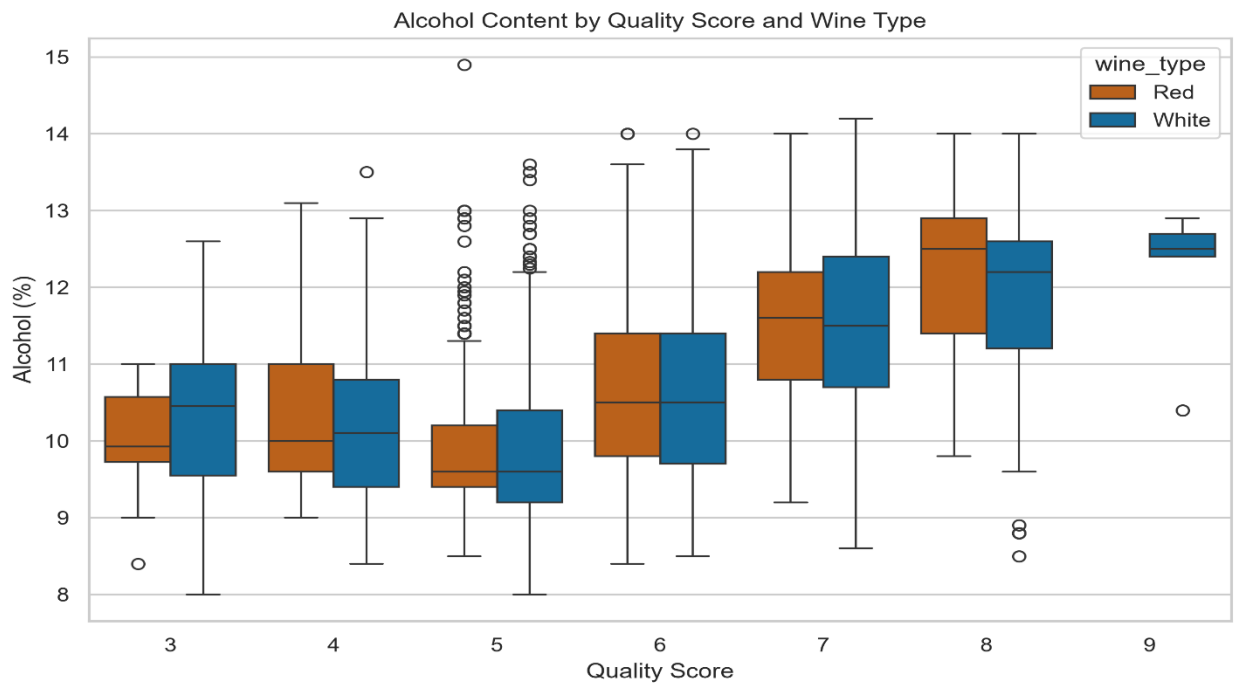
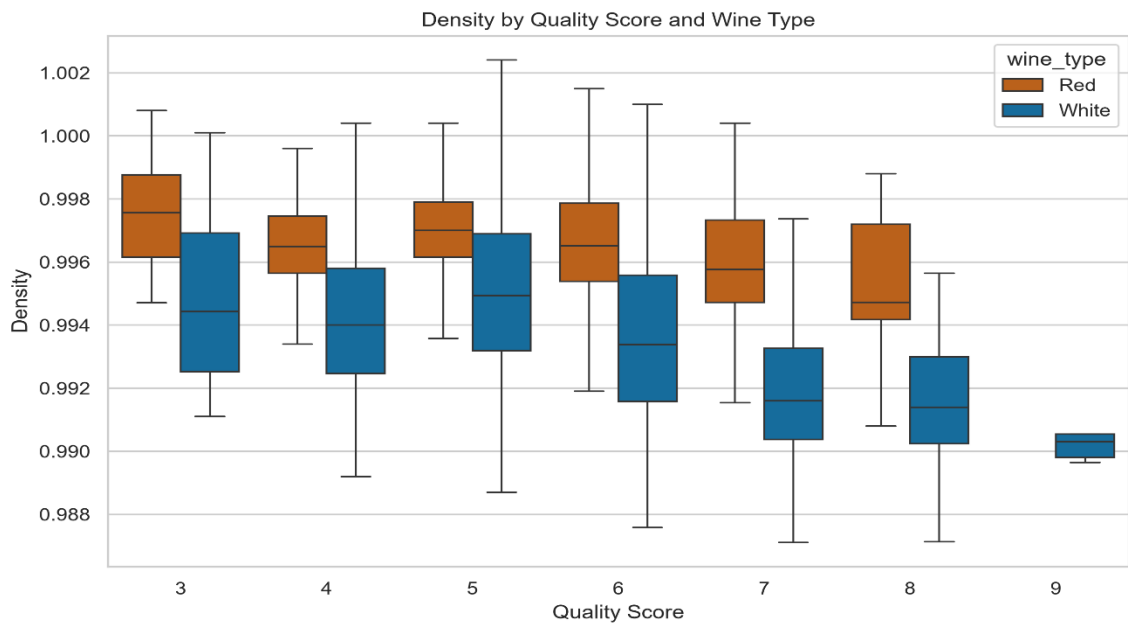
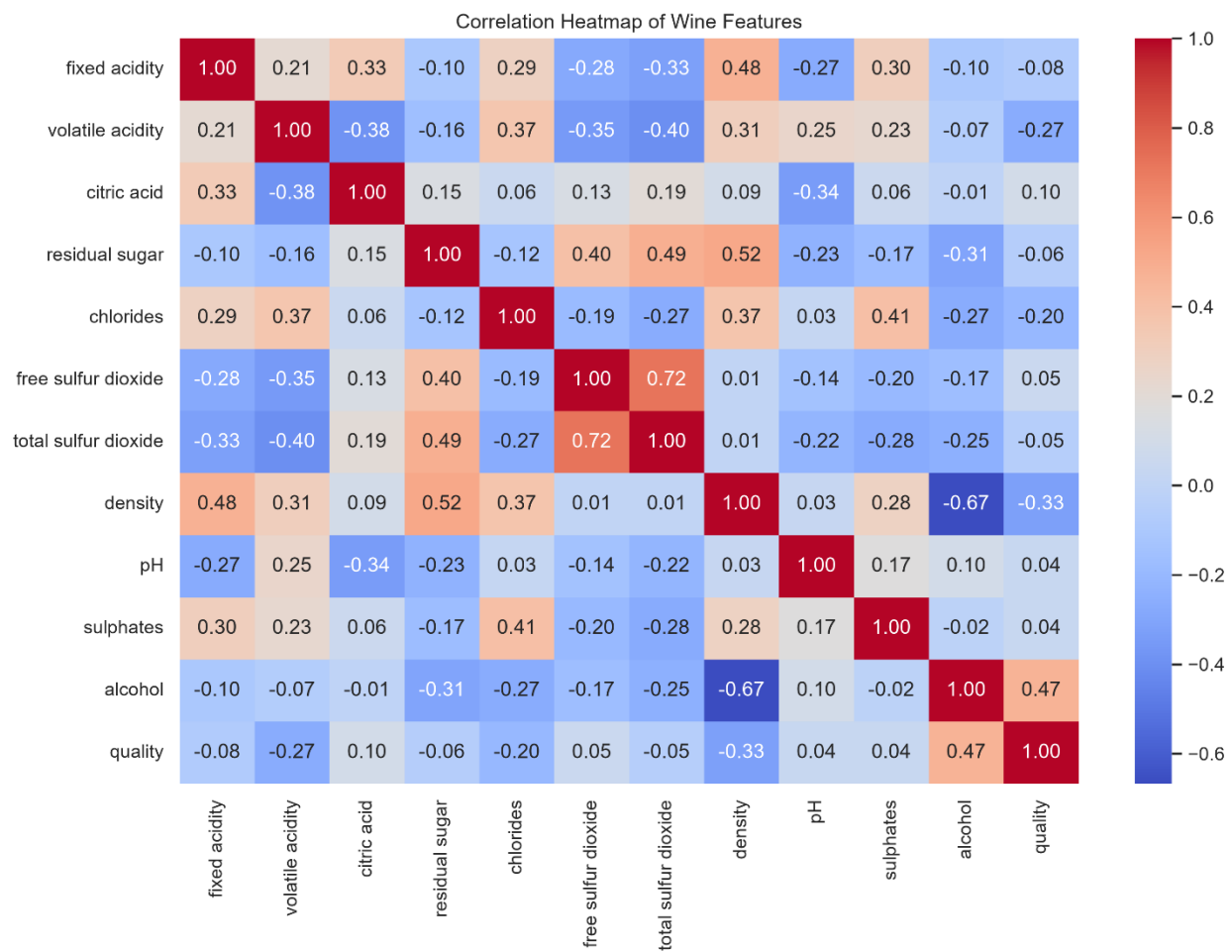


Figure 3 demonstrated an inverse relationship between density and quality. Wines receiving higher quality ratings generally exhibited lower density values, particularly among higher-rated wines.



The correlation analysis shown in **Figure 4** supported these findings. Alcohol exhibited the strongest positive correlation with quality, while density exhibited the strongest negative correlation. Although several moderate relationships were observed among predictor variables, most correlations remained relatively weak, *suggesting that wine quality is likely influenced by multiple physicochemical characteristics acting together rather than by a single dominant variable.*



Responsible Practice (Bias and Data Quality)

Several potential sources of bias and limitations should be considered.

First, wine quality ratings were assigned by experts and therefore contain an element of human judgment. Although expert ratings provide a structured assessment process, they may not fully reflect consumer preferences or generally accepted definitions of wine quality.

Second, the dataset exhibits class imbalance. Most wines received average ratings, while very low-quality and very high-quality wines were comparatively rare. This imbalance may influence conclusions drawn from the data and could potentially affect future predictive models built using the dataset.

Finally, the dataset contains only physicochemical measurements and expert quality ratings. Important contextual factors such as grape variety, production methods, region, market price, and consumer preferences were not included and may contribute to wine quality outcomes.

Reproducibility

Reproducibility was a primary consideration throughout this project. According to Danchev (2022), reproducible data science workflows should emphasize documentation, data preprocessing, exploratory analysis, modularized code, dependency management, and version control.

Several reproducibility practices were implemented throughout this project:

- The workflow was documented using Markdown explanations within a Jupyter Notebook.
- Cleaning and analysis logic was modularized into reusable Python functions.
- Software dependencies were recorded using both a requirements.txt file and a Conda environment.yml file.
- Git and GitHub were used to manage project development.
- Separate main and development branches were maintained throughout the project lifecycle.
- The notebook was successfully executed from top to bottom to verify reproducibility.

These practices allow another user to recreate both the computational environment and the analytical workflow used in this project.

References

Agarwal, A., Prabha, S., & Yadav, R. (2025). Exploratory data analysis (EDA) for banking and finance: Unveiling insights and patterns. GBS Impact: Journal of Multi Disciplinary Research. <https://doi.org/10.58419/gbs.v11i1.1112520>

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Shreffler, J., & Huecker, M. R. (2023). Exploratory data analysis: Frequencies, descriptive statistics, histograms, and boxplots. StatPearls Publishing. <https://nih.gov>.